

tests/dynamic/analyzers/ egress_neq_host_analyzer.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:**
Active self-validated analyzer (§11.4.107(10)). Signal 3 of 6 for the
dynamic data plane — real VPN egress (the hardest-to-fake
routing proof).

Companion (§11.4.18) to the in-source documentation block
at the top of the script.

Overview

Signal 3: **egress IP via the proxy $\neq$ the host's real public IP**
(§11.4.69 network_connectivity; design §13 vpn_real_egress;
research §15). The egress IP observed **through the proxy** must
equal the EXPECTED tunnel exit AND differ from the host's real
public IP. A 200 OK is **not** proof of routing; egress == host is the §15
bluff (a PASS with no VPN actually engaged). It delegates to the
committed, self-tested evidence.sh:assert_egress_ip via its
EVIDENCE_OBSERVED_IP_FILE seam, feeding a captured egress IP.

Prerequisites

- Source library tests/dynamic/lib/analyzer_common.sh + the
committed tests/lib/evidence.sh.
- awk, POSIX sh.
- A manifest (key=val):
 - observed_ip_file=egress_vpn.ip (resolved relative to the
manifest dir)
 - expected_exit=185.65.135.70
 - host_real=203.0.113.45

Usage examples

```
# Analyze a captured egress manifest:  
tests/dynamic/analyzers/egress_neq_host_analyzer.sh analyze  
    <manifest-file>
```

```
# Self-validate (golden-good PASS + golden-bad FAIL) – the default
  action:
tests/dynamic/analyzers/egress_neq_host_analyzer.sh --selftest
```

Edge cases

- **Manifest / observed-egress capture missing** → FAIL.
- **evidence.sh not found** → FAIL (cannot delegate), never a bluff.
- **egress == host_real** (no VPN diversion — the §15 bluff) → FAIL.
- **Egress is a wrong/unexpected exit** → FAIL.
- **Empty egress capture** (nothing observed) → FAIL.

§11.4.115 RED_MODE polarity

In P10 this analyzer is paired with `wg_transfer_delta`: the egress-IP echo can be cached, so the WireGuard byte-delta is the orthogonal data-plane corroboration. Consuming suites run the §11.4.115 polarity citing this analyzer as the egress oracle.

Internal behaviour

- `#!/usr/bin/env bash`, POSIX-clean (`sh -n + bash -n`, §11.4.67).
- `_egress_manifest_get` parses `key=val` with `awk`; the observed-IP file is resolved relative to the manifest dir when not absolute.
- Drives `assert_egress_ip "$proxy" "$expected_exit" "$host_real"` through the `EVIDENCE_OBSERVED_IP_FILE` seam (no network); forwards its rc.
- Self-test asserts: golden-good PASS; `egress==host`, `wrong-exit`, `empty-capture` → FAIL; `missing-manifest negative` → FAIL.

Related

- `tests/lib/evidence.sh` (`assert_egress_ip`) — the delegated oracle.
- `tests/dynamic/lib/analyzer_common.sh` — sourced base.
- Fixtures: `tests/dynamic/analyzers/fixtures/egress_neq_host/`.
- Constitution §11.4.69 / §11.4.107 / §11.4.115; design §13; research §15.

Last verified

2026-07-01 — self-test PASS (golden-good PASS + egress==host/
wrong-exit/empty FAIL); sh -n + bash -n parse-clean. Live egress
capture + WireGuard byte-delta corroboration are exercised in
P10.